

EXPERIMENT 4

Student Name: Hunar Katyal

UID: 24BDA70085

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

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Experiment 4 -- To study and implement advanced SQL queries using Joins, Correlated Query's and Conditional statement.

Aim : To study and implement correlated subqueries in SQL by retrieving employees whose salaries are among the top three highest distinct salary values within their respective departments, while correctly handling duplicate salary values using aggregate functions and joins

Question :4 Part A (i):

You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

SQL Query:

```
Select rankings, country
FROM (
  SELECT rankings, country, (
    CASE
      WHEN country = 'India' THEN 1
      WHEN country = 'Sri Lanka' THEN 2
      ELSE 3
    ) as ranks
  FROM tbl_happiness
  ORDER BY ranks, rankings
)
```

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	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	1	Finland
4	2	Denmark
5	3	Iceland
6	4	Israel
7	5	Netherlands
8	6	Sweden
9	7	Norway

Question: Part A (ii)

Make all the rankings in descending order except **India** and **Sri Lanka**

SQL Query:

```

Select rankings, country
FROM (
  SELECT rankings, country, (
    CASE
      WHEN country = 'India' THEN 1
      WHEN country = 'Sri Lanka' THEN 2
      ELSE 3
    END
  ) as ranks
FROM tbl_happiness
ORDER BY ranks ASC, rankings DESC
)

```

	rankings integer	country character varying (50)
1	126	India
2	128	Sri Lanka
3	7	Norway
4	6	Sweden
5	5	Netherlands
6	4	Israel
7	3	Iceland
8	2	Denmark
9	1	Finland

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Question: Part B

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

SQL Query:

```
select m.employee_id, m.name ,count(*), sum(e.age)/count(*) as average_age
from employees as e
JOIN employees as m
on e.reports_to = m.employee_id
group by m.employee_id, m.name
```

	employee_id [PK] integer	name character varying (50)	count bigint	average_age bigint
1	9	Hercy	2	38

Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement correlated subqueries** to solve SQL problems involving row-wise comparisons.
2. **Apply aggregate functions** such as COUNT() and DISTINCT to retrieve top-N records within groups.
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.
4. **Analyze and solve complex SQL queries** involving grouped data, ranking logic, and handling duplicate values effectively.